S.M. Abdullah

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GitHub • Blog • Portfolio Jahangirnagar University, Savar

PROFESSIONAL SUMMARY

Backend focused full-stack software developer with hands-on experience building modern web applications, AI-assisted platforms, and real-time systems. Proficient in FastAPI, React, PostgreSQL, and vector databases, with expertise in REST API development, authentication, embeddings, semantic search, and cloud deployment.

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), C++, HTML, CSS

Frameworks: FastAPI, React, Django

Databases: PostgreSQL, MySQL, SQLite, Supabase, Pinecone (vector DB)

Tools & Platforms: AWS, Git/GitHub, Docker, Postman, REST API design, pgAdmin, Crawlee, Playwright, Selenium, LangChain, n8n.

Specialties: API development, authentication flows, RAG pipelines, web crawling & scraping (Crawlee, Playwright, Selenium), LangChain, embeddings, semantic search, cloud deployment.

Soft Skills: Strong problem-solving · Critical thinking · Quick learner · Adaptability · Team collaboration

PROJECTS

BehaviorIQ – AI-Powered Behavioral Commerce Intelligence Platform *FastAPI · Python · Pinecone · PostgreSQL*

- E-commerce platforms treat every visitor the same regardless of behavior, built a FastAPI microservice that scores live session intent (0-100) from browsing signals and predicts RFM-based churn probability, turning raw pixel events into an actionable behavioral profile for every visitor
- Static product rankings ignore who's actually searching, built user behavioral vectors and a hybrid search pipeline that combines vector similarity with intent-aware reranking, so results shift based on each visitor's real-time intent score instead of a one-size-fits-all ranking
- SME owners had no way to ask questions about their own commerce data, built a RAG-powered AI assistant on top of the behavioral data layer, letting owners query intent trends, churn risk, and product performance in plain English instead of digging through dashboards

Github: Link

Ember – AI Mental Health Companion *React · FastAPI · Supabase · Hugging Face*

- Full-stack AI mental health support platform integrating Hugging Face toxicity detection models with a React frontend and FastAPI backend.
- Supabase handles persistent user data and authentication.

Live: Link | Github: Link

Onkur – AI Plant-Care App *React · FastAPI · AI/ML*

- Personal plant-care application with AI-driven daily care routines, weather-based recommendations, and image-based plant health diagnostics via Plant.id API.

Live: onkur.netlify.app | Github: Link

Universal Web Scraper & RAG Chatbot *Crawlee · Playwright · LangChain · Gemini · Pinecone*

- Built a production-grade web crawler using Crawlee + Playwright for dynamic site traversal and Trafilatura for clean content extraction.
- Implemented a full RAG pipeline: LangChain chunking → sentence-transformers embeddings → Pinecone vector store → Gemini LLM generation.
- System handles arbitrary URLs and lets users query any crawled site through a natural language interface.

Github: Link

Personal Blog Platform *FastAPI · Supabase · REST API*

- Blog platform with full CRUD, user authentication, and a responsive UI. Designed and implemented the REST API and data layer end-to-end.

Live: Link | Github: Link

EDUCATION

B.Sc. in Information and Communication Technology
Institute of Information Technology, Jahangirnagar University
CGPA: 3.82

ACHIEVEMENTS & ACTIVITIES

- Prompt Engineering Competition by AgentX — 8th Place